

BookMySeat — Requirements Document (v4)

WFO / WFH / Leave Attendance & Slot Allocation Application

Platform: OutSystems | Date: July 16, 2026 | Supersedes: v3 (July 08, 2026)

0. Summary of Requirement Changes (v3 -> v4)

#	Change	Detail
1	Teams & headcount	e_Motor — 23, Gear_Box — 20, PFG — 27 (total 70 active employees). 'Switch Gear' renamed to 'Gear_Box'.
2	Daily categories	Every working day an employee is marked one of: Work From Home (WFH), Work From Office (WFO), Leave (A).
3	Total seat capacity	63 WFO slots (unchanged). WFH and Leave are never capacity-limited.
4	No seat selection	Seat map and explicit seat picking are removed. Booking WFO auto-assigns a slot. If the employee already booked a date, the system simply shows that their slot is filled.
5	Status change (WFH <-> WFO)	Employees may change WFH to WFO (or any category) whenever they want — but not after that day is closed.
6	Closed-day edits	Editing a previous (closed) date requires manager approval; only after approval does the Edit option open, for a one-time edit. Today's date remains editable until day close.
7	Dashboard — split	Dashboard shows a split view: team-wise counts of WFO, WFH, Leave and Absent, plus overall totals and WFO slot utilization.
8	Employee addition (Manager)	New: the manager can add new employees from the Admin panel (name, work email, team). New joiners are created Active and can book immediately; headcounts and the split dashboard update automatically.

1. Application Overview

BookMySeat manages daily work-mode declarations across teams. Each employee marks one category per working day — **WFO**, **WFH** or **Leave (A)**. WFO is capacity-limited to **63 slots**; slots are allocated automatically by the system (no Book-My-Show seat grid any more). The dashboard gives managers a live, team-wise split of who is in office, at home, on leave or absent.

Core Constraints & Logic

- **Total capacity:** 63 WFO slots per working day.
- **Total workforce:** 70 active employees — PFG 27, Gear_Box 20, e_Motor 23 (7-person daily WFO deficit -> first-come, first-served).
- **Overbooking rule:** booking the 64th WFO slot triggers the “Show Full” validation and blocks the transaction. WFH / Leave are unaffected.
- **Auto slot assignment:** the employee never selects a seat. On WFO booking the server assigns the next free slot (1–63) and the UI shows “slot filled”.
- **One entry per person per date:** if a date is already booked, the UI shows the slot as filled and offers Change instead of a new booking.
- **Employee lifecycle:** new employees are added by the **manager** from the Admin panel and created as Active; leavers are deactivated (records retained for history). Headcounts above are the go-live baseline and grow as joiners are added.

2. OutSystems Data Model (Entities & Attributes)

2.1 Team (Static Entity)

Attribute	Data Type	Description
Id	Identifier	Primary Key
Label	Text	Records: PFG (27), Gear_Box (20), e_Motor (23)
Is_Active	Boolean	Soft-delete / deactivate a team

2.2 Attendance_Status (Static Entity) — NEW

Attribute	Data Type	Description
Id	Identifier	Primary Key
Label	Text	Records: WFO, WFH, Leave (A)
NeedsSlot	Boolean	True only for WFO — drives the 63-slot capacity check

2.3 Employee

Attribute	Data Type	Description
Id	Long Integer	Primary Key
Name	Text	Full name
Email	Email	Work email for notifications
TeamId	Team Identifier	FK to Team
Is_Active	Boolean	True for active; False for employees who left (history kept)

2.4 Daily_Booking (replaces Seat_Booking)

Attribute	Data Type	Description
Id	Long Integer	Primary Key
EmployeeId	Employee Identifier	FK to Employee — unique together with BookingDate
BookingDate	Date	The working day the entry applies to
StatusId	Attendance_Status Id	WFO / WFH / Leave (A)
SlotNumber	Integer (nullable)	Auto-assigned 1–63 when StatusId = WFO; null for WFH / Leave. Never chosen by the user.
CreatedOn	Date Time	When the entry was created
UpdatedOn / UpdatedBy	Date Time / Employee Id	Audit trail for changes and approved closed-day edits

2.5 Edit_Approval — NEW

Governs edits to closed (previous) dates.

Attribute	Data Type	Description
Id	Long Integer	Primary Key
EmployeeId	Employee Identifier	Who requested the edit
BookingDate	Date	The closed date to be edited
Reason	Text	Optional justification

Attribute	Data Type	Description
Status	Static (Pending / Approved / Rejected / Used)	Approval workflow state
RequestedOn / DecidedOn	Date Time	Timestamps
ApprovedBy	Employee Identifier	The manager who decided

2.6 Holiday

Unchanged: Id, HolidayDate (Date), Name (Text), CreatedBy (Employee Identifier).

3. Workflow Descriptions

3.1 Daily Booking Flow (no seat selection)

- 1 Employee opens “My status”, picks a date (today or up to 7 days ahead) and a category: WFO / WFH / Leave (A).
- 2 If the employee already has an entry for that date, the UI shows the slot as **filled** with the current category — no re-booking; a Change action is offered instead.
- 3 **Server action (WFO only):** aggregate counts Daily_Booking records where BookingDate = selected date and StatusId = WFO.
- 4 **Validation decision:** Count ≥ 63 -> raise “Show Full” and block. Count < 63 -> create the record and auto-assign SlotNumber = next free slot. The employee never picks a seat.
- 5 WFH and Leave (A) entries are created without any capacity check or slot.

3.2 Status Change Flow (WFH <-> WFO)

- 1 Employees can change their category (e.g. WFH -> WFO) whenever they want for **today or any future date** — but not after that day is closed.
- 2 Changing **to WFO** re-runs the capacity check; if 63 slots are filled the change is blocked with “Show Full”.
- 3 Changing **away from WFO** releases the auto-assigned slot immediately, making it available to others.
- 4 Today's entry remains editable until **day close** (the 6:00 PM close-out timer / end of day). After close the date locks.

3.3 Closed-Day Edit Flow (manager approval)

- 1 For a previous (closed) date the Edit option is locked; the employee taps “**Request manager approval**” -> an Edit_Approval record is created (Pending).
- 2 The manager sees the request in Admin -> Edit approvals and Approves or Rejects it.
- 3 On **Approve**, the Edit option opens for that employee & date for a **one-time edit**; saving the edit marks the approval as Used and re-locks the date.
- 4 On **Reject** the date stays locked. All approved edits are audit-trailed (UpdatedBy / UpdatedOn).

3.4 Dashboard Analytics Flow — Split View

- **Team-wise split:** aggregates group Daily_Booking by Employee.TeamId and StatusId to show, per team (PFG / Gear_Box / e_Motor): WFO, WFH, Leave (A) and Absent counts. Absent = active members with no entry for the date.
- **Overall split:** totals across teams — WFO x/63, WFH, Leave, Absent — plus a WFO slot-utilization ring (Occupied = WFO count; Empty = 63 - count) with the Show Full indicator.
- **Trend:** WFO slots filled per day for the next 8 days for capacity planning.

3.5 Daily Automated Timer (6:00 PM)

- **Day close-out:** locks the current date's entries (editable only via manager-approved edit thereafter).

- **Notification:** sends the split summary (team-wise WFO / WFH / Leave / Absent) and next-day reminders via Email / Microsoft Teams webhook.

3.6 Employee Addition Flow (Manager) — NEW

- 1 Manager opens Admin -> **Add new employee** and enters Full name, Work email and Team (PFG / Gear_Box / e_Motor).
- 2 **Validations:** all three fields are mandatory; the email must be a valid format and unique across all employee records (active and left).
- 3 On save, the Employee record is created with **Is_Active = True**; the new joiner can book WFO / WFH / Leave (A) immediately and receives a welcome notification by email.
- 4 Team headcounts, the split dashboard and the Excel report include the new employee automatically from the moment of creation.
- 5 Access control: the Add employee screen and its server action are restricted to the Manager role — employees never see it.

4. Roles & Permissions

Capability	Employee	Manager
Book daily category — WFO (auto slot) / WFH / Leave (A)	Yes	Yes
Change own category for today / future dates (before day close)	Yes	Yes
Request edit of a closed (previous) date	Yes	Yes
Approve / reject closed-day edit requests	No	Yes
View split dashboard (team-wise WFO / WFH / Leave / Absent)	Yes	Yes
Add new employee (name, email, team — created Active)	No	Yes
Deactivate an employee who left (records kept for history)	No	Yes
Shut down a date as company holiday	No	Yes
Export split attendance report (Excel)	No	Yes

5. Calendar & Booking Rules

- **Weekends:** Saturday and Sunday are never bookable — disabled automatically.
- **Company holidays:** the manager can shut down any date; entries on it are cancelled and members notified.
- **Show Full:** when all 63 WFO slots for a date are taken, further WFO bookings (and changes to WFO) are blocked with “Show Full”. WFH / Leave stay open.
- **One entry per person per date:** a booked date shows as “slot filled”; the employee changes it rather than booking again.
- **Booking window:** up to 7 days in advance (configurable).
- **Edit window:** today and future dates — free self-service edits until day close; previous dates — locked, manager approval required (one-time edit per approval).

6. Dashboard — Split (Team-wise Counts)

The dashboard is a split view for a selected date (default today):

Widget	Content
Overall split cards	WFO (slots filled x/63 with remaining), WFH, Leave (A), Absent / not marked — across all 70 employees
WFO utilization ring	Occupied vs 63; turns red with a “Show Full” badge at capacity

Widget	Content
Team-wise split bars	Per team (PFG 27 / Gear_Box 20 / e_Motor 23): stacked WFO · WFH · Leave · Absent vs headcount
Split table	Team × {Members, WFO, WFH, Leave (A), Absent} with a totals row
Trend chart	WFO slots filled per day, next 8 days

7. Excel Export — Split Report (Manager Only)

- One-click Export to Excel (.xlsx) from the Admin panel.
- One row per employee — name, team, status, then per working day one of **WFO / WFH / L / A**, plus per-person totals of each category.
- Team-wise totals block: WFO / WFH / Leave / Absent per team for the reporting period.
- Access control: button and server action restricted to the Manager role.

8. Validation Summary

- 1 Weekend selected -> "Weekends are not bookable."
- 2 Holiday selected -> "This date is shut down by the manager: {holiday name}."
- 3 WFO capacity reached (64th booking or change to WFO) -> "Show Full — all 63 WFO slots are booked."
- 4 Date already booked -> slot shown as **filled** with current category; duplicate booking blocked — "Already marked {category} for this date."
- 5 No category selected -> "Pick a category first — WFO, WFH or Leave (A)."
- 6 Edit attempt on a closed (previous) date without approval -> blocked — "Day closed. Request manager approval to edit."
- 7 Edit request already pending -> "Approval request already sent."
- 8 Approved edit is one-time: saving consumes the approval and re-locks the date.
- 9 Same-day edits allowed only until day close (6:00 PM close-out).
- 10 Concurrent WFO bookings: capacity re-checked server-side at commit; the later transaction gets "Show Full".
- 11 Admin: holiday without date/name, or duplicate date -> blocked with message.
- 12 Admin — add employee: missing name, email or team -> blocked with message; invalid email format -> "That doesn't look like a valid email address"; duplicate email -> "An employee with this email already exists."

9. Retained / Deferred Features

- **Removed in v4:** seat map UI, seat selection, seat transfer/swap, favorite seats, premium tiers, seat-hold timer, waiting list (WFH replaces the queue when WFO is full).
- **Retained:** holidays, weekend lock, role split, notifications (booking / change / approval decisions / 6 PM digest), employee lifecycle (manager adds joiners, deactivates leavers), Excel export, 7-day window.
- **Deferred (future):** configurable capacity beyond the fixed 63, recurring patterns, calendar sync, team quotas, guest booking.